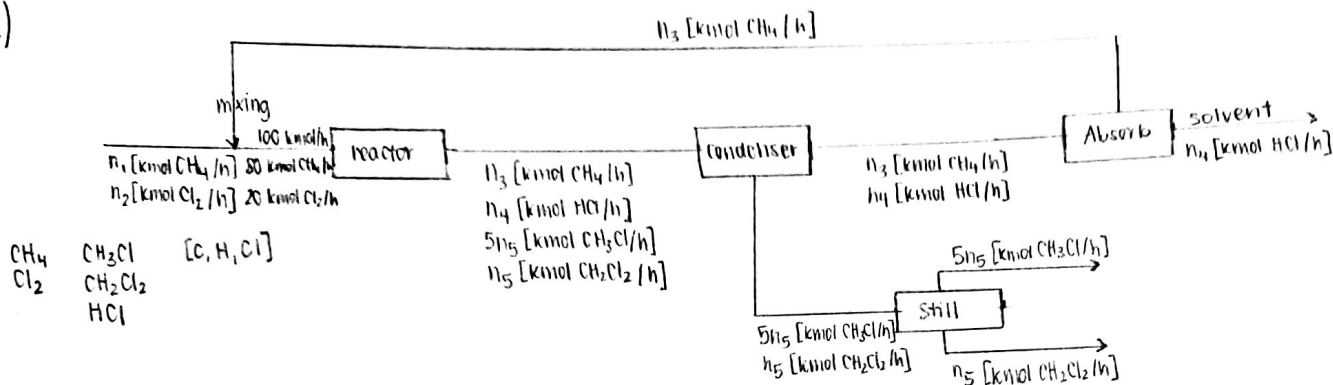


CHE 210 Homework 08

1)



a) DOF overall

unknowns: n_1, n_2, n_4, n_5 (4) equations: C, H, Cl (3) $DOF = 4 - 3 = 1$ DOF mixing unknowns: n_1, n_2, n_3 (3) equations: CH_4, Cl_2 (2) $DOF = 3 - 2 = 1$ DOF reactor unknowns: n_3, n_4, n_5 (3) equations: C, H, Cl (3) $DOF = 3 - 3 = 0$

Equations Mixing

$$\begin{array}{ll} CH_4 \text{ balance} & Cl_2 \text{ balance} \\ n_1 + n_3 = 80 & n_2 = 20 \end{array}$$

$$\begin{array}{l} \text{Equations Reactor} \\ \text{Carbon balance} \\ 80 = n_3 + 5n_5 + n_5 \end{array}$$

DOF condenser

unknowns: n_3, n_4, n_5 (3)

DOF absorber

unknowns: n_3, n_4 (2)

DOF still

unknowns: n_4, n_5 (2)

$$\begin{array}{l} \text{Hydrogen balance} \\ 4 \cdot 80 = 4n_3 + n_4 + 3 \cdot 5n_5 + 2 \cdot n_5 \end{array}$$

Chlorine balance

$$2 \cdot 20 = n_4 + 5n_5 + 2 \cdot n_5$$

b)

$$\begin{array}{l} CH_4 \text{ balance} \\ Cl_2 \text{ balance} \\ C \text{ balance} \\ H \text{ balance} \\ Cl \text{ balance} \end{array} \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 6 \\ 0 & 0 & 4 & 1 & 17 \\ 0 & 0 & 0 & 1 & 7 \end{bmatrix} \begin{bmatrix} n_1 \\ n_2 \\ n_3 \\ n_4 \\ n_5 \end{bmatrix} = \begin{bmatrix} 80 \\ 20 \\ 80 \\ 320 \\ 40 \end{bmatrix}$$

$$n_1 = 17.14 \text{ kmol } CH_4/h$$

$$n_2 = 20 \text{ kmol } Cl_2/h$$

$$n_3 = 62.86 \text{ kmol } CH_4/h$$

$$n_4 = 20 \text{ kmol } HCl/h$$

$$n_5 = 2.86 \text{ kmol } CH_2Cl_2/h \rightarrow 5n_5 = 14.29 \text{ kmol } CH_3Cl/h$$

c)

$$CH_3Cl: 12.01 + 3(1.008) + 35.45 = 50.48 \text{ kg } CH_3Cl / \text{kmol}$$

$$1000 \frac{\text{kg } CH_3Cl}{h} \times \frac{1 \text{ kmol } CH_3Cl}{50.48 \text{ kg } CH_3Cl} = 19.81 \frac{\text{kmol } CH_3Cl}{h}$$

$$\text{Scale Factor} = \frac{19.81 \frac{\text{kmol } CH_3Cl}{h}}{14.29 \frac{\text{kmol } CH_3Cl}{h}} = 1.39$$

$$n'_1 = n_1 \cdot \text{Scale Factor} = 17.14 \text{ kmol } CH_4/h \cdot 1.39 = 23.82 \text{ kmol } CH_4/h$$

$$n'_2 = n_2 \cdot \text{Scale Factor} = 20 \text{ kmol } Cl_2/h \cdot 1.39 = 27.80 \text{ kmol } Cl_2/h$$

$$n'_3 = n_3 \cdot \text{Scale Factor} = 62.86 \text{ kmol } CH_4/h \cdot 1.39 = 87.38 \text{ kmol } CH_4/h \rightarrow \text{recycle stream}$$

$$\text{Fresh Feed} = n'_1 + n'_2 = 23.82 + 27.80 \text{ kmol} = 51.62 \text{ kmol Fresh Feed/h}$$

$$x_1 = \frac{23.82 \text{ kmol } CH_4/h}{51.62 \text{ kmol Fresh Feed/h}} = 0.4614 \text{ } CH_4 \rightarrow 46.14 \text{ mol } \% \text{ } CH_4$$

$$x_2 = \frac{27.80 \text{ kmol } Cl_2/h}{51.62 \text{ kmol Fresh Feed/h}} = 0.5386 \text{ } Cl_2 \rightarrow 53.86 \text{ mol } \% \text{ } Cl_2$$

$$\frac{40}{40}$$

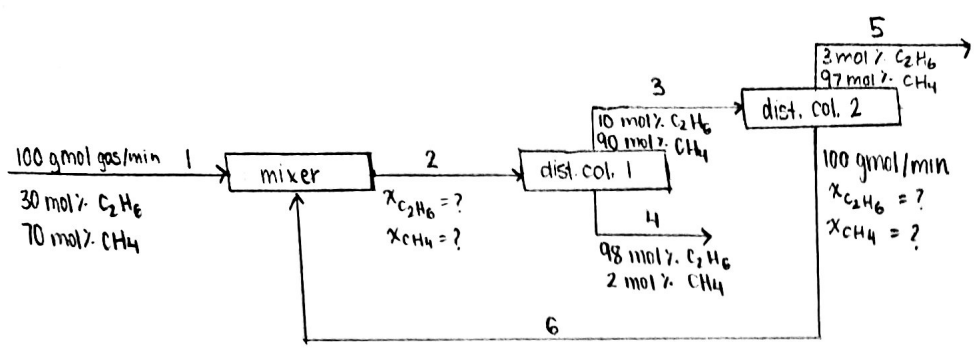
$$1) \quad 40$$

$$2) \quad 30$$

$$3) \quad 30$$

$$\frac{100}{100}$$

2)



Mixer Species Balance: C_2H_6

$$x_{1,C_2H_6} \cdot n_1 + x_{6,C_2H_6} \cdot n_6 = x_{2,C_2H_6} \cdot n_2$$
$$0.30 \cdot 100 + 100 x_{6,C_2H_6} = n_2 x_{2,C_2H_6}$$
$$30 + 100 x_{6,C_2H_6} = n_2 x_{2,C_2H_6}$$

Dist. Col. 1 Species Balance: C_2H_6

$$x_{2,C_2H_6} \cdot n_2 = x_{3,C_2H_6} \cdot n_3 + x_{4,C_2H_6} \cdot n_4$$
$$n_2 x_{2,C_2H_6} = 0.10 n_3 + 0.98 n_4$$

Dist Col 2. Species Balance: C_2H_6

$$x_{3,C_2H_6} \cdot n_3 = x_{5,C_2H_6} \cdot n_5 + x_{6,C_2H_6} \cdot n_6$$
$$0.10 n_3 = 0.03 n_5 + 100 x_{6,C_2H_6}$$

Mole Fraction Equations

$$x_{2,C_2H_6} + x_{2,CH_4} = 1$$
$$x_{6,C_2H_6} + x_{6,CH_4} = 1$$

C_2H_6 sum

$$30 + 100 x_{6,C_2H_6} = n_2 x_{2,C_2H_6}$$
$$n_2 x_{2,C_2H_6} = 0.10 n_3 + 0.98 n_4$$
$$+ 0.10 n_3 = 0.03 n_5 + 100 x_{6,C_2H_6}$$
$$30 = 0.98 n_4 + 0.03 n_5$$

Dist. Col. 2 Molar Flow Balance

$$n_3 = n_5 + n_6$$
$$n_3 = 71.58 + 100 \quad \checkmark$$
$$n_3 = 171.58 \text{ gmol/min}$$

Dist. Col. 2 Mol %: C_2H_6

$$0.10 n_3 = 0.03 n_5 + 100 x_{6,C_2H_6}$$
$$0.10 \cdot 171.58 = 0.03 \cdot 71.58 + 100 x_{6,C_2H_6}$$
$$x_{6,C_2H_6} = 0.15$$

Dist. Col 1 Molar Flow Balance

$$n_2 = n_3 + n_4$$
$$n_2 = 171.58 + 28.42 \quad \checkmark$$
$$n_2 = 200 \text{ gmol/min}$$

Dist. Col 1 Mol %: C_2H_6

$$n_2 x_{2,C_2H_6} = 0.10 n_3 + 0.98 n_4$$
$$200 x_{2,C_2H_6} = 0.10 \cdot 171.58 + 0.98 \cdot 28.42$$
$$x_{2,C_2H_6} = 0.23 \quad \checkmark$$

Mixer Species Balance: CH_4

$$x_{1,CH_4} \cdot n_1 + x_{6,CH_4} \cdot n_6 = x_{2,CH_4} \cdot n_2$$
$$0.70 \cdot 100 + 100 x_{6,CH_4} = n_2 x_{2,CH_4}$$
$$70 + 100 x_{6,CH_4} = n_2 x_{2,CH_4}$$

Dist. Col. 1 Species Balance: CH_4

$$x_{2,CH_4} \cdot n_2 = x_{3,CH_4} \cdot n_3 + x_{4,CH_4} \cdot n_4$$
$$n_2 x_{2,CH_4} = 0.90 n_3 + 0.02 n_4$$

Dist. Col. 2 Species Balance: CH_4

$$x_{3,CH_4} \cdot n_3 = x_{5,CH_4} \cdot n_5 + x_{6,CH_4} \cdot n_6$$
$$0.90 n_3 = 0.97 n_5 + 100 x_{6,CH_4}$$

CH_4 sum

$$70 + 100 x_{6,CH_4} = n_2 x_{2,CH_4}$$
$$n_2 x_{2,CH_4} = 0.90 n_3 + 0.02 n_4$$
$$0.90 n_3 = 0.97 n_5 + 100 x_{6,CH_4}$$
$$70 = 0.02 n_4 + 0.97 n_5$$

$$\begin{bmatrix} 0.98 & 0.03 \\ 0.02 & 0.97 \end{bmatrix} \begin{bmatrix} n_4 \\ n_5 \end{bmatrix} = \begin{bmatrix} 30 \\ 70 \end{bmatrix} \rightarrow \begin{matrix} n_4 = 28.42 \text{ gmol/min} \\ n_5 = 71.58 \text{ gmol/min} \end{matrix}$$

Dist. Col. 2 Mol %: CH_4

$$0.90 n_3 = 0.97 n_5 + 100 x_{6,CH_4}$$
$$0.90 \cdot 171.58 = 0.97 \cdot 71.58 + 100 x_{6,CH_4}$$
$$x_{6,CH_4} = 0.85 \quad \checkmark$$

Dist Col 1 Mol %: CH_4

$$n_2 x_{2,CH_4} = 0.90 n_3 + 0.02 n_4$$
$$200 x_{2,CH_4} = 0.90 \cdot 171.58 + 0.02 \cdot 28.42$$
$$x_{2,CH_4} = 0.77 \quad \checkmark$$

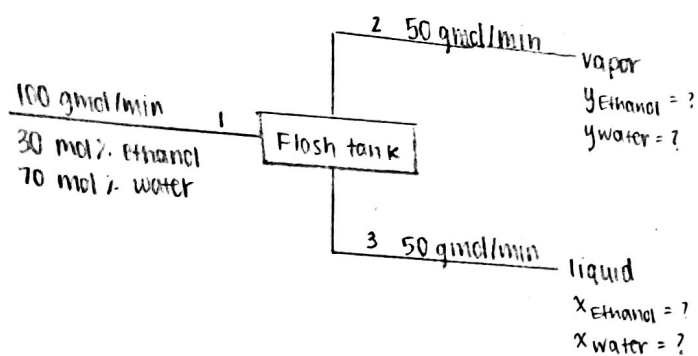
Fractional Recovery: C_2H_6

$$f_{R,C_2H_6} = \frac{x_{4,C_2H_6} \cdot n_4}{x_{1,C_2H_6} \cdot n_1} = \frac{0.98 \cdot 28.42}{0.30 \cdot 100} = 0.928$$

Fractional Recovery: CH_4

$$f_{R,CH_4} = \frac{x_{5,CH_4} \cdot n_5}{x_{1,CH_4} \cdot n_1} = \frac{0.97 \cdot 71.58}{0.70 \cdot 100} = 0.992 \quad \checkmark$$

30/30



Ethanol balance:

$$z_{\text{ethanol}} \cdot n_1 = y_{\text{ethanol}} \cdot n_2 + x_{\text{ethanol}} \cdot n_3$$

$$0.30 \cdot 100 = 50 y_{\text{ethanol}} + 50 x_{\text{ethanol}}$$

$$30 = 50 y_{\text{ethanol}} + 50 x_{\text{ethanol}}$$

Mole Fraction

$$x_{\text{ethanol}} + x_{\text{water}} = 1$$

$$y_{\text{ethanol}} + y_{\text{water}} = 1$$

At 85.3°C

$$x_{\text{ethanol}} = 0.1238 \quad x_{\text{water}} = 0.8762$$

$$y_{\text{ethanol}} = 0.4704 \quad y_{\text{water}} = 0.5296$$

Ethanol

$$30 = 50 \cdot 0.4704 + 50 \cdot 0.1238$$

$$30 = 29.71 \text{ [Approximate]}$$

Water

$$70 = 50 \cdot 0.5296 + 50 \cdot 0.8762$$

$$70 = 70.29 \text{ [Approximate]}$$

Fractional Recovery: Ethanol and Water

$$f_{\text{rec, ethanol}} = \frac{y_{\text{ethanol}} \cdot n_2}{z_{\text{ethanol}} \cdot n_1} = \frac{0.4704 \cdot 50}{0.30 \cdot 100} = 0.784 \text{ [Ethanol]}$$

$$f_{\text{rec, water}} = \frac{x_{\text{water}} \cdot n_3}{z_{\text{ethanol}} \cdot n_1} = \frac{0.8762 \cdot 50}{0.70 \cdot 100} = 0.626 \text{ [water]}$$

Separation factor,

$$\alpha_{\text{ethanol, water}} = \frac{y_{\text{ethanol}} \cdot x_{\text{water}}}{x_{\text{ethanol}} \cdot y_{\text{water}}} = \frac{0.4704 \cdot 0.8762}{0.1238 \cdot 0.5296} = 6.29$$

$$\frac{30}{30}$$